

Theory Concepts — Part A

Conceptual Understanding, House Price Regression Project (PR.1)

1. What are Supervised Learning Algorithms?

Supervised learning algorithms learn a mapping from input features (X) to a known output/target (y) using a labeled training dataset. The model is trained by comparing its predictions to the true labels and adjusting itself to minimize the error, so it can later predict labels for unseen data. Linear Regression, Logistic Regression, Decision Trees, and SVMs are common examples.

2. Difference between Regression Algorithms vs Classification Algorithms

Regression algorithms predict a continuous numeric value (e.g., house price, temperature), and are evaluated with error metrics like MSE / RMSE / R^2 . Classification algorithms predict a discrete category or class label (e.g., spam vs not-spam), and are evaluated with metrics like accuracy, precision, recall, and F1-score. The core difference is the type of target variable — continuous vs categorical.

3. Explain Simple Linear Regression

Simple Linear Regression models the relationship between one independent variable (X) and one dependent variable (y) as a straight line:

$$y = \beta_0 + \beta_1 \cdot X + \varepsilon$$

where β_0 is the intercept, β_1 is the slope (change in y per unit change in X), and ε is the random error term. The coefficients are estimated by minimizing the Sum of Squared Errors (Ordinary Least Squares) between actual and predicted values.

4. Assumptions of Linear Regression

- **Linearity** — the relationship between X and y is linear.
- **Independence of errors** — residuals are not correlated with each other (no autocorrelation).
- **Homoscedasticity** — residuals have constant variance across all predicted values (no funnel/cone shape).
- **Normality of residuals** — the errors are approximately normally distributed, which matters for valid confidence intervals and hypothesis tests.
- **No (or low) multicollinearity** — in multiple regression, independent variables shouldn't be highly correlated with each other, or coefficient estimates become unstable.

5. What is Bias–Variance Trade-Off?

Bias is the error from overly simplistic assumptions in the model — a high-bias model underfits and misses real patterns. Variance is the error from excessive sensitivity to the training data — a high-variance model overfits, capturing noise as if it were signal. Total expected error is approximately Bias-squared plus Variance plus irreducible noise. As model complexity increases, bias tends to fall while variance rises — the trade-off is finding the complexity level that minimizes total error on unseen data, not just training error.

6. Overfitting and Underfitting (with examples)

- **Underfitting:** the model is too simple to capture the underlying pattern — e.g., fitting a straight line to a clearly curved relationship. Both training and test error are high.
- **Overfitting:** the model is too complex and starts memorizing noise in the training data — e.g., fitting a degree-15 polynomial through 20 points. Training error is very low, but test error is high because the model doesn't generalize.
- In this project's notebook, Parts F and H show this directly: a high-degree polynomial fits training points only marginally better, with no real test-set benefit — an early sign of overfitting risk, while the single-feature models overall show underfitting relative to Multiple Regression.